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UNIVERSITY
NAVI MUMBAI

Semester - III | **B. Tech. Computer Engineering** | **39724303**

2413CEC3T3 | **Discrete Mathematics**

Date: 27/11/2025 | **Time: 10:00 AM TO 01:00 PM** | **MaxMarks:100**

Q.1	**Attempt all questions**	**20 Marks**	**CO**	**BT**
a)	Show that $(p \leftrightarrow q) \equiv [(p \to q) \wedge (q \longrightarrow p)]$	4	CO1	BT2
b)	Let $A = \{1,4,5\}$ and R be a relation on A defined by aRb if $a + b \leq 6$. Write R and check for reflexivity and transitivity.	4	CO2	BT3
c)	A box contains 6 red and 4 blue marbles. Two marbles are drawn at random without replacement. What is the probability that the first marble drawn is red and the second marble drawn is blue?	4	CO3	BT2
d)	State handshaking Lemma. Also if the graph has vertices of degree 0, 2,2, 3 and 9. How many edges the graph have?	4	CO4	BT1
e)	Show that Klein group $K_4 = \{e, a, b, c\}$ is a group with respect on operation defined on K_4.	4	CO5	BT2

Q.2	**Attempt Any 4**	**20 Marks**	**CO**	**BT**
a)	Check whether the following statement is Tautology, Contradiction or Contingency $[(p \vee q) \wedge (\sim p \vee r)] \longrightarrow (q \vee r)$	5	CO1	BT3
b)	Determine whether the relation R on the set of integers is reflexive symmetric and antisymmetric, where aRb if and only if $ab \geq 0$	5	CO2	BT5
c)	A test for a certain disease gives a positive result with 95% probability if a person has the disease and a positive result with 5% probability if a person does not have the disease. If 0.1% of the population has the disease, find the probability that a randomly tested person who receives a positive result actually has the disease.	5	CO3	BT1
d)	Define the following types of graphs and also give examples: i. Bipartite graph ii) Regular graph iii) complete graph	5	CO4	BT1
e)	Let $G = U(8) = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ and let $*$ be a binary operation defined on G as $a * b = ab(mod\ 8)$. Is $(G,*)$ a cyclic group?	5	CO5	BT5
f)	Show that (2,5) encoding function $e: B^2 \to B^5$ defined by $e(00) = 00000$, $e(01) = 01110$, $e(10) = 10101$, $e(11) = 11011$ is a group code.	5	CO6	BT2

Q.3. **Attempt Any 4** — **20 Marks** — **CO** — **BT**

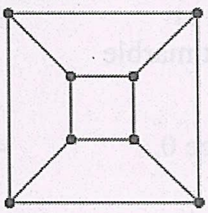
a) Obtain the conjunctive normal form of the following — 5 — CO1 — BT1

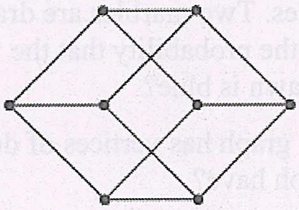
$$[q \vee (p \wedge r)] \wedge \sim [(p \vee r) \wedge q]$$

b) $Let\ f: R - \left\{\frac{1}{2}\right\} \rightarrow R - \left\{\frac{3}{2}\right\}\ defined\ by\ f(x) = \frac{3x+5}{2x-1}$ show that f is bijective also find inverse of f. — 5 — CO2 — BT1

c) There are 2 bags. The first bag contains 3 white and 4 black balls. The second bag contains 2 white and 3 black balls. A ball is chosen at random from the first bag and transferred to second bag. Then a ball is chosen at random from second bag. What is the probability that the ball is white? — 5 — CO3 — BT2

d) Define Isomorphism of graph and check whether the following graphs are isomorphic or not. — 5 — CO4 — BT4





e) Let $G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle| \ a, b, c, d\ are\ integers \right\}$. Check whether G is a group with respect to matrix multiplication. — 5 — CO5 — BT3

f) Consider the parity matrix H given by $H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Determine the group code $e_H: B^2 \rightarrow B^5$. — 5 — CO6 — BT5

Q.4. **Attempt Any 2** — **20 Marks** — **CO** — **BT**

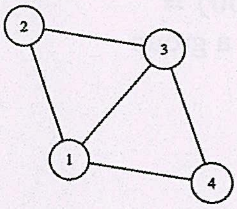
a) Are the following arguments are valid? — 10 — CO1 — BT

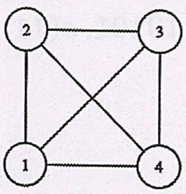
a) $p \vee q$
$\sim q$
$\therefore p$

b) x is positive or x is negative.
If x is positive, then $x^2 > 0$.
If x is negative, then $x^2 > 0$.
$\therefore x^2 > 0$.

b) If A={1, 2, 3, 5, 7}then show that the relation $aRb\ if\ a \leq b$ is partial order relation and hence draw Hasse Diagram. — 10 — CO2 — BT3

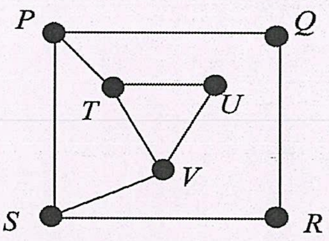
c) Define spanning tree and construct all non-isomorphic spanning trees for the following graphs. — 10 — CO4 — BT4





Q.5. Attempt Any 2 **20 Marks** **CO** **BT**

a) Define Eulerian graph and Hamiltonian graph with example. Consider the graph G below, 10 CO4 BT3



(i) Is G Eulerian? Either state an Euler circuit on G or explain why G is not Eulerian.
(ii) Is G Hamiltonian? Either state a Hamiltonian cycle on G or explain why G is not Hamiltonian.

b) Let $G = \mathbb{Z}$ be a group with the operation $*$ defined as $a * b = a + b$. Determine whether the following are subgroups of G. 10 CO5 BT5

i. $H = 2\mathbb{Z}$
ii. $H = 2\mathbb{Z} \cup 3\mathbb{Z}$.

c) Check whether the $(2, 6)$ encoding function $e: B^2 \longrightarrow B^6$ defined by $e(00) = 000000, e(01) = 111001, e(10) = 101011$ and $e(11) = 010010$ is a group code? Also find the minimum distance of e and the number of errors e can detect. 10 CO6 BT2